

University of Prishtina “Hasan PRISHTINA”



**Faculty of Electrical and Computer Engineering
Master in Computer and Software Engineering, 2025-2026**

Course: Data Preparation and Visualization

Week III – Data Preprocessing

Dr. Sc. Mërgim H. HOTI

What is today's agenda?

➤ Today we are going to learn about:

- Data understanding;
- Data preprocessing.

What is Data?

- Collection of data objects and their attributes;
- An attribute is a property or characteristic of an object:
 - Examples: eye color of a person, temperature, etc;
 - Attributes is also known as variable, field characteristic, or feature.
- A collection of attributes describe an object:
 - Object is also known as record, point, case, sample, entity, or instance.

Objects

Attributes				
Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Attribute Values

- Attribute values are numbers or symbols assigned to an attribute;

- Difference between attributes and attribute values:
 - Same attribute can be mapped to different attribute values:
 - Example: height can be measured in feet or meters.

 - Different attributes can be mapped to the same set of values:
 - Example: Attribute values for ID and age are integers;
 - But properties of attributes values can be different:
 - ID has “no limit”, but age has a maximum and minimum value.

Discrete and Continuous Attributes

➤ Discrete Attribute:

- Has only a finite or countably infinite set of values;
- Examples: zip codes, counts, or the set of words in a collection of documents;
- **Note:** binary attributes are a special case of discrete attributes.

➤ Continuous Attribute:

- Has real numbers as attribute values;
- **Examples:** temperature, height, or weight;
- Practically, real values can only be measured with limited precision;
- Continuous attributes are typically represented as floating-point variables.

Type of Attributes

➤ Categorical (Qualitative):

▪ Nominal:

- Examples: ID numbers, eye color, zip codes, etc;

▪ Ordinal:

- Examples: ranking (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}.

➤ Numeric (Quantitative):

▪ Interval:

- Examples: temperatures in Celsius or Fahrenheit;

▪ Ratio:

- Examples:
 - **Height:** 0 cm means no height. A person who is 180 cm tall is twice as tall as someone who is 90 cm.
 - **Weight:** 0 kg means no weight. A 10 kg object weighs twice as much as a 5 kg object.
 - **Age:** 0 years means no age. Someone who is 40 years old is twice as old as someone who is 20 years old.

Type of Attributes

Nominal	Ordinal	Interval	Ratio
No order or rank, Equality, Categories, No mathematical meaning	Order, Rank, No equal distance between 2 ranks	No meaningful zero, Equal distances between 2 points	True zero
Zip code, Gender, Color, Ethnics, Political affiliations, Religion affiliations, Major field, Nationality, Martial status, Sports player's back numbers, AM & PM, Dates, Credit card numbers.	Grade (678910), Judging (1 st , 2 nd , 3 rd , 4 th), Rating scale (excellent, good, bad), Ranking of sports players, Week, Months, etc.	STA score, IQ, Temperature, 12 hours of day, Date of week, Days of a month, Months of a year.	Height, Weight, Time, Salary, Age, 24 hours of days (0 = 24).

Nominal: Arlis is young and Eraza is Old.

Ordinal: Arlis is younger then Eraza.

Interval: Arlis is 20 years younger than Eraza.

Ratio: Arlis is twice as young as Eraza.

Types of Data Sets

➤ Record:

- *Data Matrix;*
- *Document Data;*
- *Transaction Data.*

➤ Graph:

- *World Wide Web;*
- *Molecular Structures.*

➤ Ordered:

- *Spatial Data;*
- *Temporal Data;*
- *Sequential Data;*
- *Genetic Sequence Data.*

Record Data



Data that consists of a collection of records, each of which consists of a fixed set of attributes:

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multidimensional space, where each dimension represents a distinct attribute;
- Such dataset can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute.

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Document Data



- Each document becomes a “term” vector:
- each term is a component (attribute) of the vector;
 - the value of each component is the number of times the corresponding term occurs in the document.

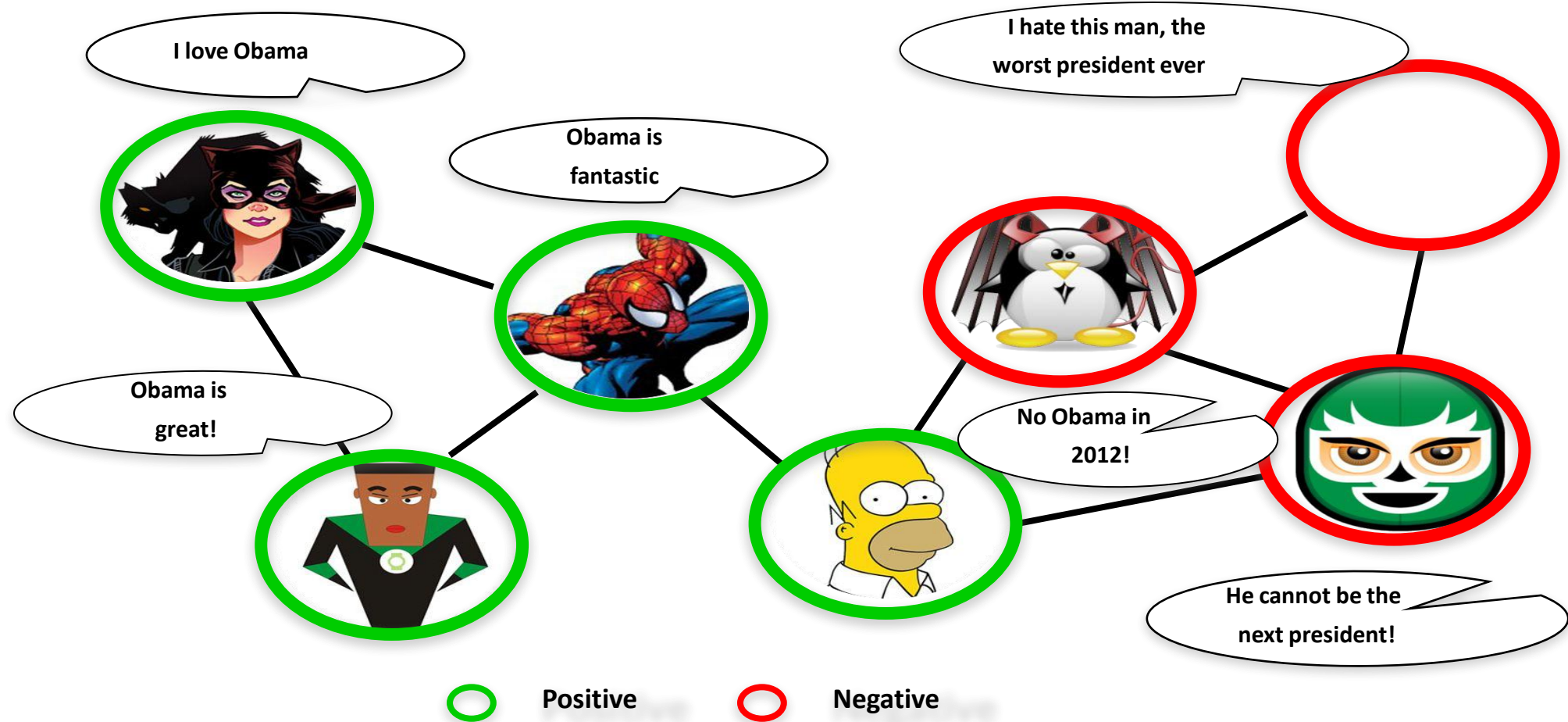
	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Transaction Data

- A special type of record data, where:
- Each record (transaction) involves a set of items;
 - For example, consider a grocery store:
 - The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

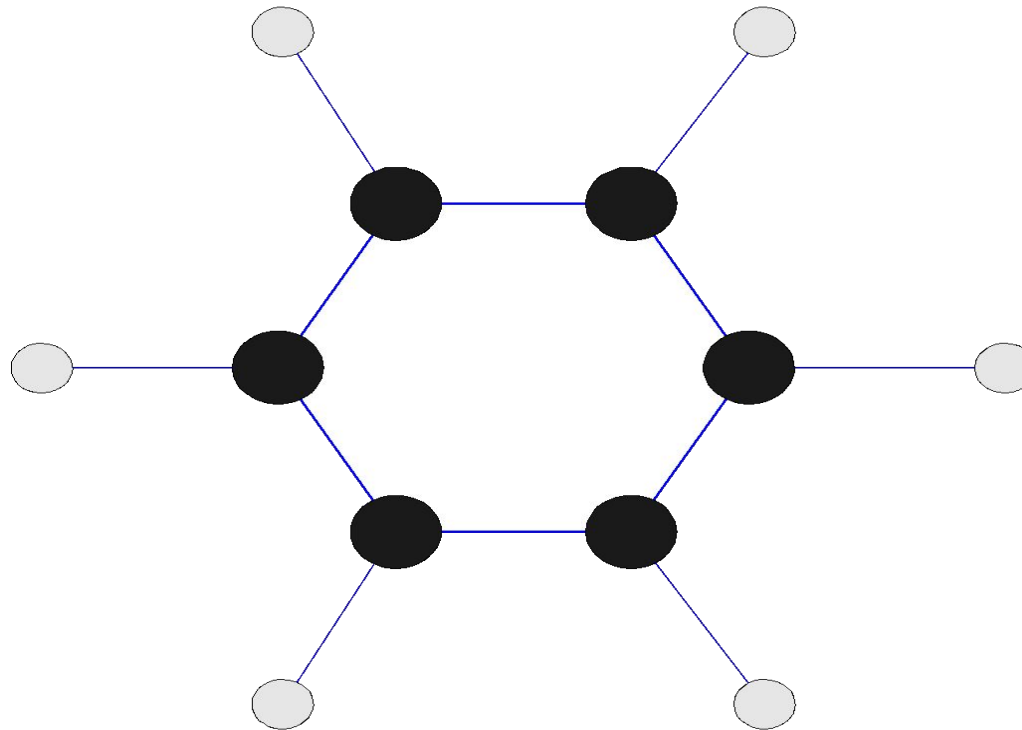
Graph Data: “Love Obama”



Chemical Data

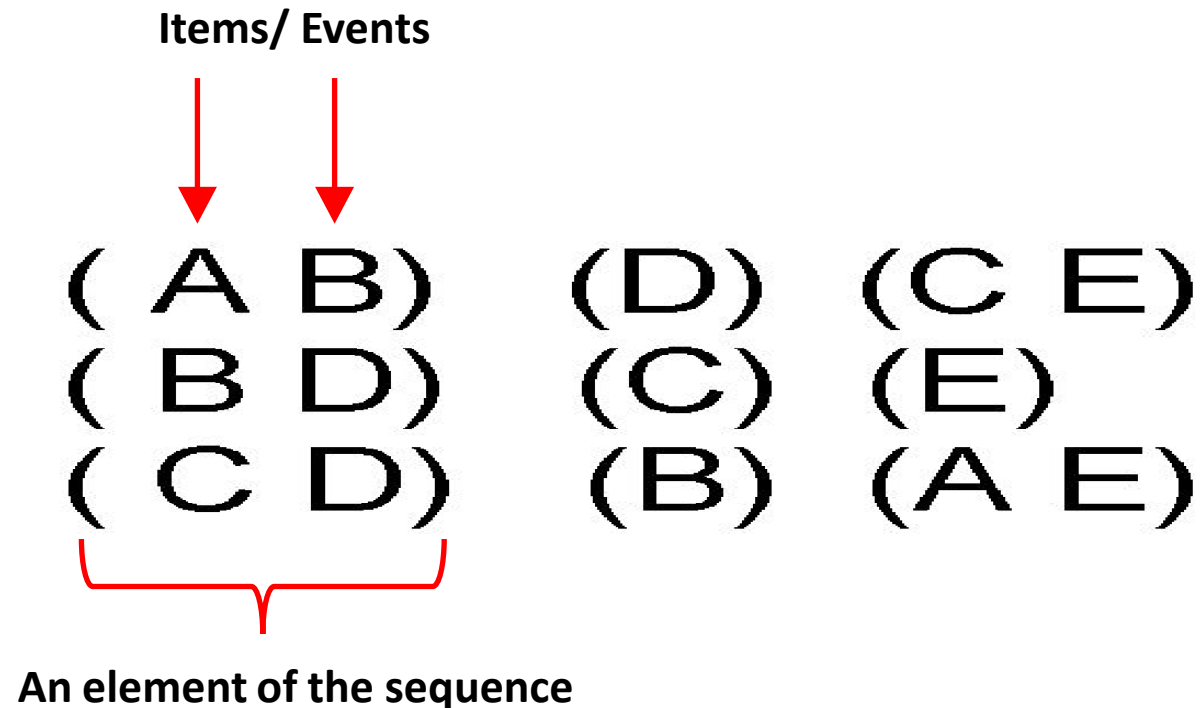


➤ Benzene Molecule: C_6H_6



Ordered Data

➤ Sequence of transactions

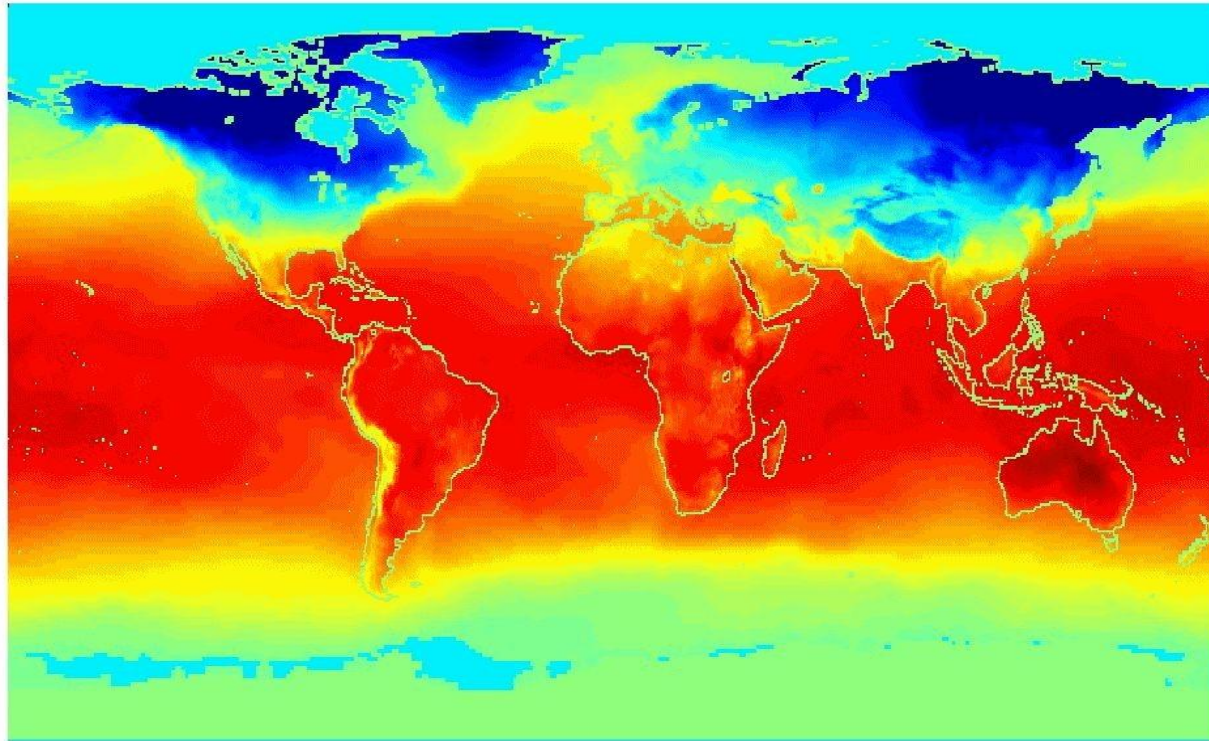


Ordered Data



➤ Spatio-Temporal Data

Jan



Data Quality

- What kinds of data quality problems?
- How can we detect problems with the data?
- What can we do about these problems?

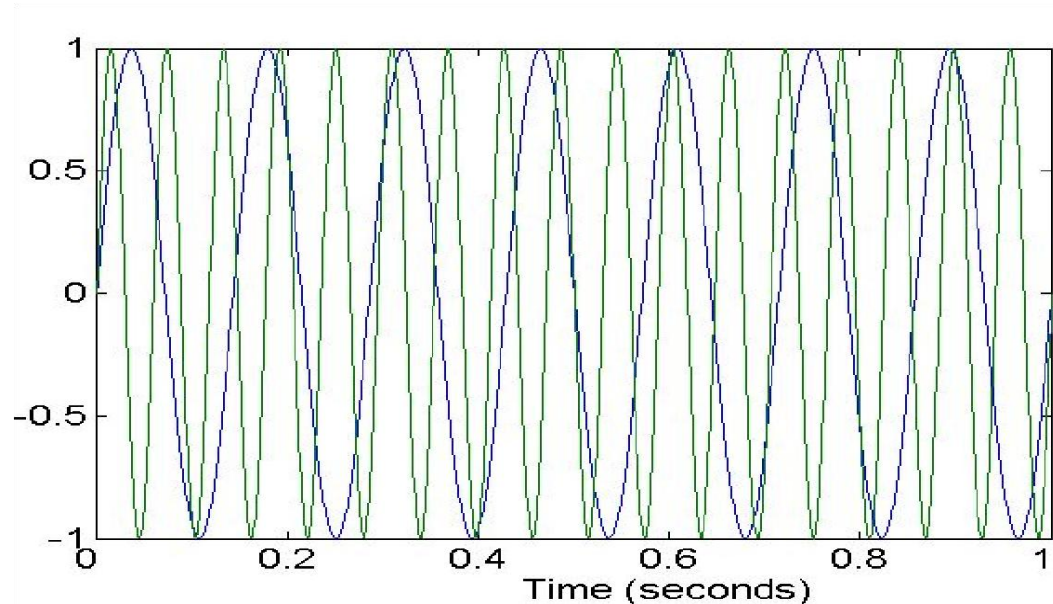
- Examples of data quality problems:
 - Noise and outliers;
 - Missing values;
 - Duplicate data.

Noise!

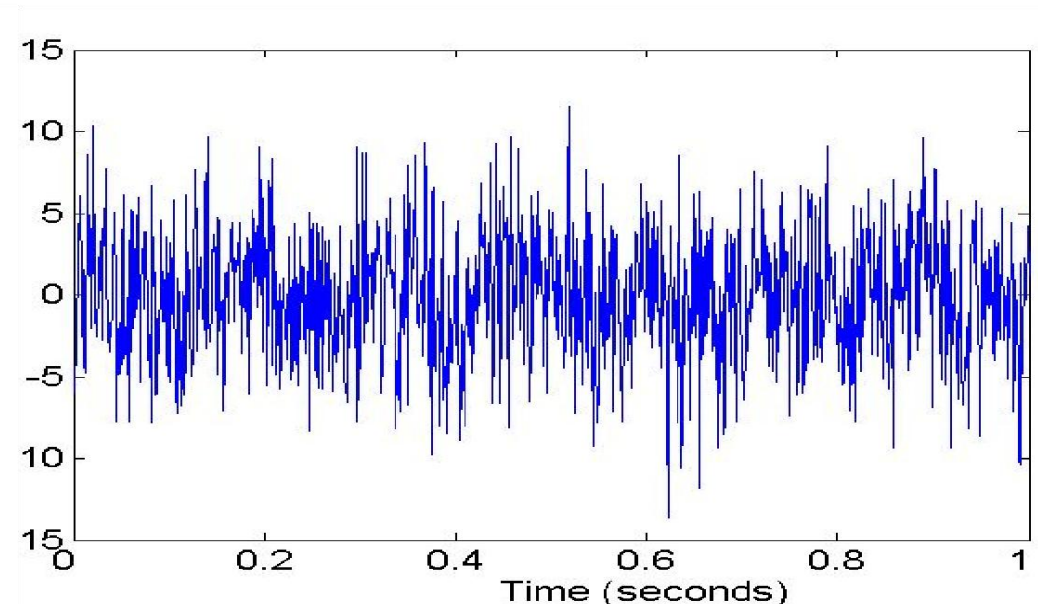


➤ **Noise refers to modification of original values**

➤ Examples: distortion of a person's voice when talking on a poor phone and “snow” on television screen.



Two Sine Waves



Two Sine Waves + Noise

Noise!

- **Noise** in data means unwanted or random changes in the original values that make the data less accurate or reliable.
- Think of it like “static” or “distortion” - it doesn’t destroy the data completely, but it makes it harder to interpret correctly:
 - **Examples:**
 - In *audio*, it’s the crackling sound in a poor phone call;
 - In *images*, it’s the “snowy” or grainy dots on a photo;
 - In *data*, it can be wrong sensor readings, typing mistakes, or random fluctuations in measurements.
- Why it matters?
 - **Noise** can confuse algorithms and make patterns harder to detect. That’s why during data preprocessing, we often:
 - **Filter** out noise (e.g., smoothing time series);
 - **Detect** outliers, or;
 - **Clean** corrupted entries.

Outliers!



- Outliers are data objects with characteristics that are considerably different than most of the other data objects in the data set.



Missing Values

➤ Reasons for missing values:

- Information is **not collected** (for example: people decline to give their age and weight);
- Attributes may **not be applicable to all cases** (for example: annual income is not applicable to children).

➤ Handling missing values:

- *Eliminate* Data Objects;
- *Estimate* Missing Values;
- *Ignore* the Missing Value During Analysis;
- *Replace* with all possible values (weighted by their probabilities).

Duplicate Data

- **Dataset may include data objects that are duplicates, or almost duplicates of one another:**
 - Major issue when merging data from heterogeneous sources.

- **Examples:**
 - Same person with multiple email address.

- **Data cleaning:**
 - Process of dealing when duplicate data issues.

Data Preprocessing

1. Aggregation;
2. Sampling;
3. Dimensionality reduction;
4. Feature subset selection;
5. Discretization and binarization;
6. Attribute transformation.

Aggregation

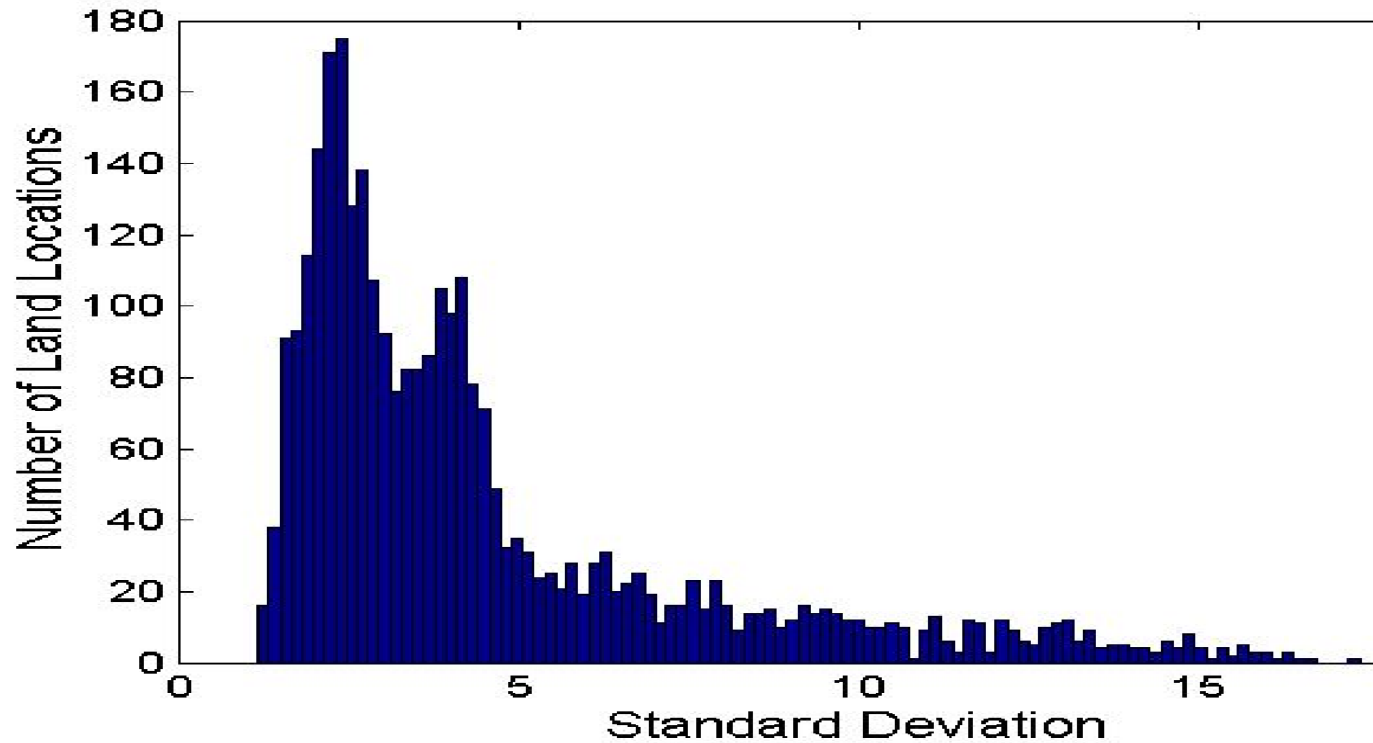
- Combining two or more attributes (for objects) into a single attribute (or object).

- Purpose:
 - Data reduction:
 - Reduce the number of attributes or objects.
 - Change of scale:
 - Cities aggregated into regions, states, countries, etc.
 - More “stable” data:
 - Aggregated data tends to have less variability.

Aggregation

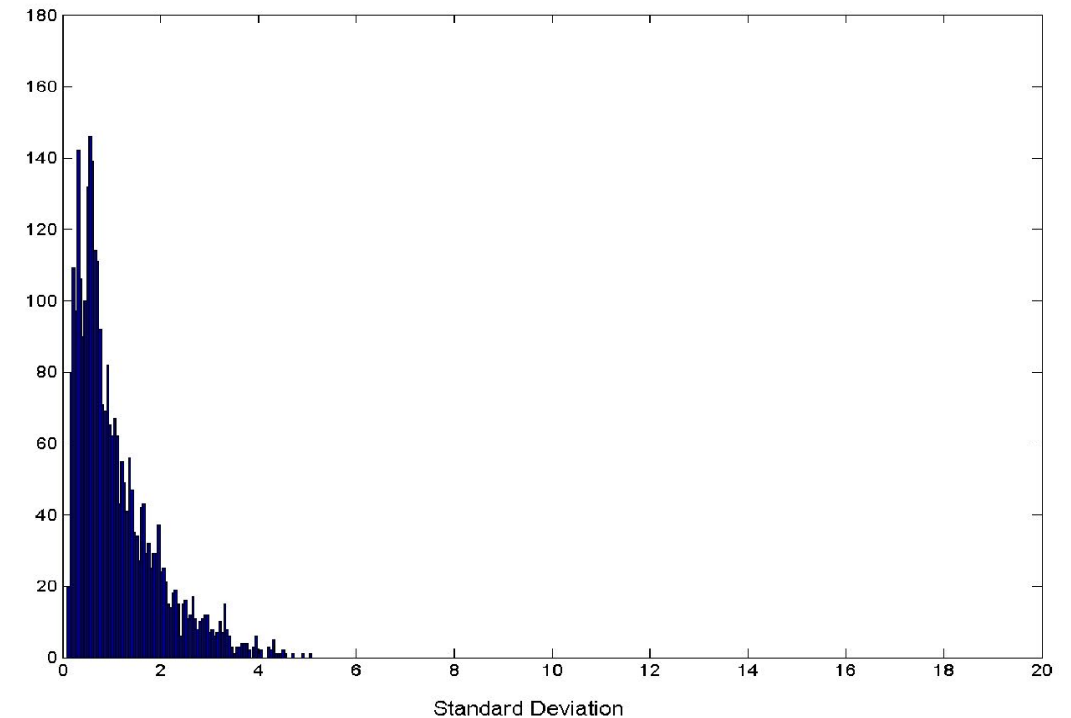


Variation of Precipitation in Australia



Standard Deviation of Average

Monthly Precipitation



Standard Deviation of Average

Yearly Precipitation

Sampling



- Sampling is the main technique employed for data selection:
 - It is often used for both the preliminary investigation of the data and the final data analysis.
- Statisticians sample because **obtaining** the entire set of data of interest is too expensive or time consuming;
- Sampling is used in data mining because **processing** the entire set of data of interest is too expensive or time consuming.

Sampling



➤ The key principle for effective sampling is following:

- Using a sample will work almost as well as using the entire data sets, if the sample is representative.
- A sample is representative if it has approximately the same property (of interest) as the original set of data .

Types of Sampling

- Simple Random Sampling (SRS):
 - There is an equal probability of selecting any particular item.
- Sampling without replacement:
 - As each item is selected, it is removed from the population.
- Sampling with replacement:
 - Objects are not removed from the population as they are selected for the sample:
 - In sampling with replacement, the same object can be picked up more than once.
- Stratified sampling:
 - Split the data into several partitions, then draw random samples from each partition.

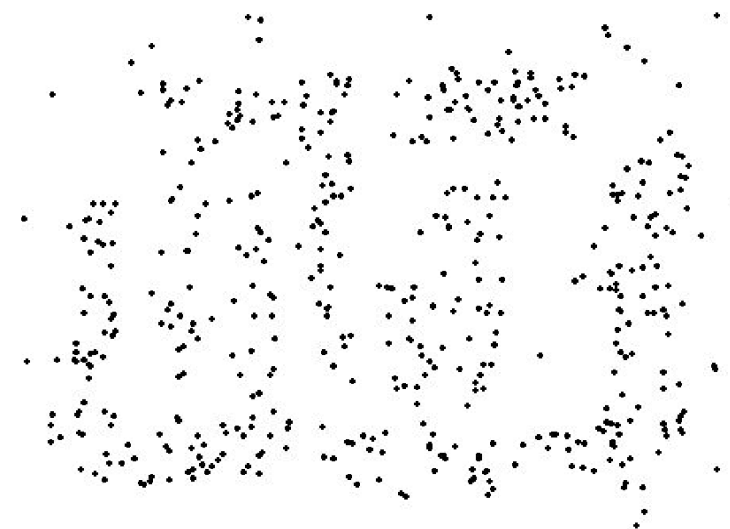
Sample Size



8000 points



2000 points



500 points

Curse of Dimensionality

- When dimensionality increases, data becomes increasingly sparse in the space that it occupies;
- Definitions of density and distance between points, which is critical for clustering and outliers' detection, become less meaningful.

Dimensionality Reduction

➤ Purpose:

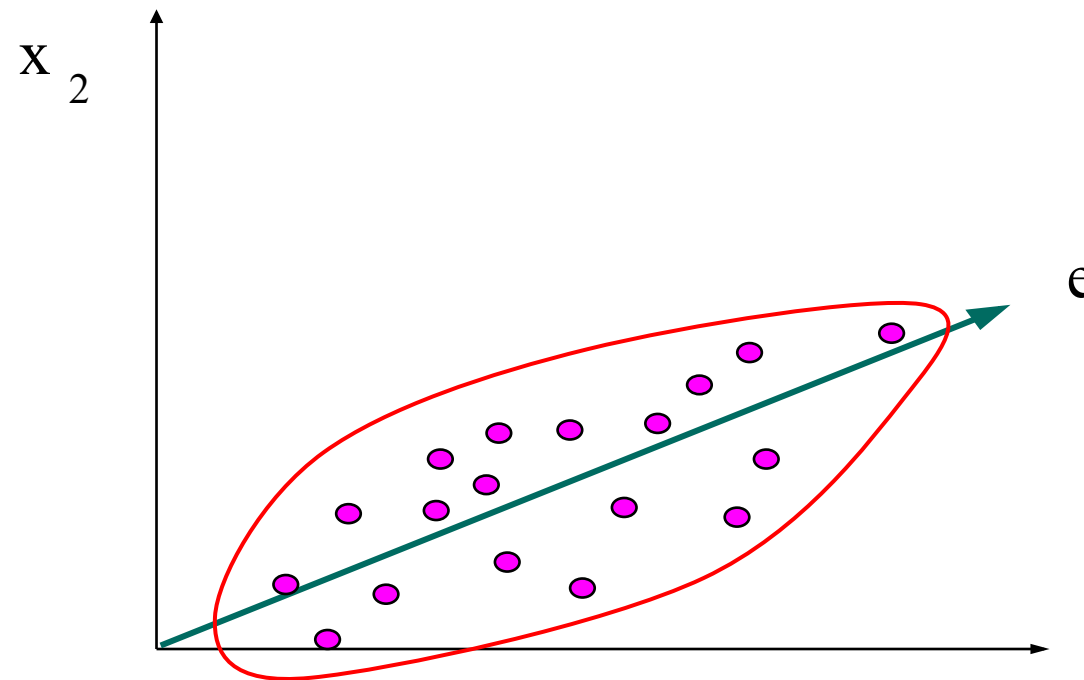
- Avoid curse of dimensionality;
- Reduce amount of time and memory required by data mining algorithms;
- Allow data to be more easily visualized;
- May help to eliminate irrelevant features or reduce noise.

➤ Techniques:

- Principal Component Analysis;
- Singular Value Decomposition;
- Other: supervised and non-linear techniques.

Dimensionality Reduction: PCA

➤ Goal is to find a projection that captures the largest amount of variation in data.



Feature Subset Selection

- Another way to reduce dimensionality of data:

- **Redundant** features:
 - Duplicate much or all of the information contained in one or more other attributes;
 - Example: purchase price of a products and the amount of sales tax paid.

- **Irrelevant** features:
 - Contain no information that is useful for the data analysis task at hand;
 - Example: students' ID is often irrelevant to the task of predicting students' GPA.

Feature Subset Selection

➤ Techniques:

➤ Brute-force approach:

- Try all possible feature subsets as input to data mining algorithm;

➤ Embedded approaches:

- Feature selection occurs naturally as part of the data mining algorithms.

➤ Filter approaches:

- Features are selected before data mining algorithms is run.

Class work



➤ Objective:

- To apply key data preprocessing operations - including data inspection, cleaning, and aggregation - on a dataset of the student's choice, enhancing its quality and analytical readiness.

➤ Task Description:

- Each student (or group) will select a dataset of their own, from any domain or topic, and conduct a short, structured data preprocessing workflow. The primary goal is to **identify data quality issues**, apply **appropriate correction techniques**, and **summarize the dataset** through aggregation.

➤ Instructions:

- Step 1: Data Exploration
- **Import and inspect** your dataset;
- **Identify** the structure and characteristics of the data, including:
 - Number of records (rows) and attributes (columns);
 - Attribute names and data types (numeric, categorical, date/time, etc.);
- Detect potential data quality issues, such as:
 - **Missing** or inconsistent values;
 - **Duplicate** records;
 - **Noisy** or outlier observations.
- Document your observations in short notes or comments within your code or worksheet.
- **Goal:** Develop an initial understanding of the dataset's structure, content, and attribute characteristics to guide further preprocessing steps.

Questions?

Dr. Sc. Mërgim H. HOTI

Faculty of Electrical and Computer Engineering,
University of Prishtina “Hasan PRISHTINA”,
Email: mergim.hoti@uni-pr.edu